

Parameter monotonicity and a sharp threshold for Nguyen's special entire function

Candidate full-solution packet

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Status: candidate full proof likely valid. The paper's open parameter-monotonicity conjecture has a positive answer. Consequently membership is governed by one threshold a_0 , and an exact rational certificate gives $3.9642 < a_0 < 3.9643$. The argument uses the paper's Theorem 1, but not its erroneous numerical Theorem 2(ii).

1 Question and result

For $a > 1$ let

$$F_a(z) = \sum_{k=0}^{\infty} \frac{z^k}{\prod_{j=1}^k (a^j + 1)}.$$

Nguyen proved that $F_a \in \mathcal{LP}$ if and only if there is an $x \in (a+1, a^2+1)$ with $F_a(-x) \leq 0$ [1, Theorem 1]. The paper then conjectured that membership is monotone in the parameter: if $F_{a_1} \in \mathcal{LP}$ and $a_2 > a_1$, must $F_{a_2} \in \mathcal{LP}$?

Theorem 1 (Positive answer to the conjecture). *If $1 < a_1 < a_2$ and $F_{a_1} \in \mathcal{LP}$, then $F_{a_2} \in \mathcal{LP}$. More precisely, set*

$$q_2(a) = \frac{a^2 + 1}{a + 1}, \quad H_a(y) = F_a(-(a+1)y).$$

For every $a \geq (3 + \sqrt{17})/2$ and $0 < y \leq q_2(a)$,

$$\frac{\partial}{\partial a} H_a(y) < 0.$$

2 Proof Intuition

The normalized variable y is the key point. It freezes the left endpoint of the sign-detecting interval at 1, while $q_2(a)$ is increasing. Thus a single negative witness for a_1 remains admissible and becomes strictly more negative for every larger parameter. Analytically, differentiation in a produces an alternating series. Pairing each even term with its following odd term leaves a negative pair: the exceptional first pair reduces to one explicit positive polynomial, while every later pair is dominated by the exponential growth of $a^{k+1} + 1$. This is the monotonicity mechanism.

3 Proof of parameter monotonicity

Write

$$H_a(y) = \sum_{k=0}^{\infty} (-1)^k A_k(a) y^k, \quad A_k(a) = \frac{(a+1)^k}{\prod_{j=1}^k (a^j + 1)}.$$

Here $A_0 = A_1 = 1$. For $j \geq 2$ define

$$\delta_j(a) = \frac{ja^{j-1}}{a^j + 1} - \frac{1}{a + 1}, \quad \lambda_k(a) = \sum_{j=2}^k \delta_j(a).$$

Since

$$ja^{j-1}(a+1) - (a^j + 1) = (j-1)a^j + ja^{j-1} - 1 > 0,$$

all δ_j and λ_k are positive. Logarithmic differentiation gives $A'_k = -\lambda_k A_k$ for $k \geq 2$. Termwise differentiation is legitimate by locally uniform absolute convergence, and pairing consecutive terms gives

$$\partial_a H_a(y) = \sum_{\substack{k \geq 2 \\ k \text{ even}}} \left(-\lambda_k A_k y^k + \lambda_{k+1} A_{k+1} y^{k+1} \right). \quad (1)$$

We show that every parenthesis is negative. Because

$$\frac{A_{k+1}}{A_k} = \frac{a+1}{a^{k+1}+1} \quad \text{and} \quad y \leq q_2(a),$$

it is enough to prove

$$\frac{\lambda_{k+1}}{\lambda_k} < \frac{a^{k+1}+1}{a^2+1}. \quad (2)$$

The first pair, $k = 2$, is the only close one. Direct simplification shows that (2) is equivalent to $P(a) > 0$, where

$$P(a) = a^7 - 2a^6 - 4a^5 + 2a^4 - 6a^3 + a^2 - a + 1.$$

For $a = u + 7/2$ its coefficients, from constant term upward, are

$$\frac{90771}{128}, \frac{236319}{64}, \frac{156727}{32}, \frac{49107}{16}, \frac{8521}{8}, \frac{845}{4}, \frac{45}{2}, 1.$$

Hence $P(a) > 0$ for $a \geq 7/2$.

For $k \geq 4$, observe that

$$\delta_2 \geq \frac{1}{a+1}, \quad \delta_{k+1} < \frac{k+1}{a}.$$

Consequently

$$\frac{\lambda_{k+1}}{\lambda_k} = 1 + \frac{\delta_{k+1}}{\lambda_k} \leq 1 + \frac{\delta_{k+1}}{\delta_2} < 2k + 3.$$

For $a \geq 3$ we also have

$$\frac{a^{k+1}+1}{a^2+1} > 2k+3 \quad (k \geq 4).$$

Indeed the case $k = 4$ reduces to $a^5 - 11a^2 - 10 > 0$, true at $a = 3$ and thereafter by differentiation; and the left side more than doubles when k is increased by one, which proves the assertion by induction. This proves (2) for every pair in (1), and therefore $\partial_a H_a(y) < 0$ whenever $a \geq 7/2$ and $0 < y \leq q_2(a)$.

It remains only to place a possible Laguerre–Pólya parameter in this range. The standard coefficient estimate reproduced as Lemma 1 in [1] says that, for this increasing sequence of second quotients, $F_a \in \mathcal{LP}$ forces $q_2(a) \geq 3$. Thus

$$a \geq \frac{3 + \sqrt{17}}{2} > \frac{7}{2}.$$

Now suppose $F_{a_1} \in \mathcal{LP}$. The sign criterion supplies $y_0 \in (1, q_2(a_1))$ with $H_{a_1}(y_0) \leq 0$. Since q_2 is strictly increasing, $y_0 < q_2(a)$ throughout $[a_1, a_2]$. The inequality just proved therefore gives

$$H_{a_2}(y_0) < H_{a_1}(y_0) \leq 0.$$

Applying the sign criterion once more yields $F_{a_2} \in \mathcal{LP}$, proving Theorem 1.

4 Existence and exact localization of the threshold

Corollary 2. *There is a number $a_0 > 1$ such that*

$$F_a \in \mathcal{LP} \iff a \geq a_0.$$

Moreover,

$$\boxed{3.9642 < a_0 < 3.9643}.$$

Proof. Theorem 1 makes the membership set upward closed, and it is nonempty by Hutchinson's $q_n \geq 4$ criterion. It is also closed. Indeed, if a_n decreases to its infimum and $F_{a_n} \in \mathcal{LP}$, choose sign witnesses $y_n \in (1, q_2(a_n))$. A subsequence converges to $y \in [1, q_2(a_0)]$, and continuity gives $H_{a_0}(y) \leq 0$. The endpoints cannot occur: $H_a(1) > 0$ by the alternating-series test, while at $y = q_2(a)$ the first three terms sum to 1 and the remaining alternating tail is larger than $-q_2/q_3 > -1$. Hence $1 < y < q_2(a_0)$, and the sign criterion gives $F_{a_0} \in \mathcal{LP}$.

For the numerical enclosure, all arithmetic below is exact. Put

$$a_- = \frac{19821}{5000} = 3.9642, \quad S_{5,a}(x) = \sum_{k=0}^5 \frac{(-1)^k x^k}{\prod_{j=1}^k (a^j + 1)}.$$

Map each of the 128 equal subintervals of $[a_- + 1, a_-^2 + 1]$ to $[0, 1]$ and express S_{5,a_-} in the degree-five Bernstein basis. The accompanying verifier checks all $128 \cdot 6 = 768$ rational coefficients; every one is positive, and the least is approximately $4.6399567291958566 \cdot 10^{-6}$. Thus $S_{5,a_-} > 0$ throughout the interval. The remaining alternating tail begins with a positive term and has strictly decreasing magnitudes, so $F_{a_-}(-x) > S_{5,a_-}(x) > 0$ there. The sign criterion implies $F_{a_-} \notin \mathcal{LP}$.

At the upper rational endpoint

$$a_+ = \frac{39643}{10000} = 3.9643, \quad x_+ = \frac{2}{3}(a_+^2 + 1),$$

the same verifier obtains

$$S_{6,a_+}(x_+) = -\frac{N_+}{D_+} \approx -8.43053729684106 \cdot 10^{-6} < 0,$$

where the verifier prints the positive integers N_+ and D_+ in full. The tail after the sixth section begins negative and has decreasing magnitudes, hence $F_{a_+}(-x_+) < S_{6,a_+}(x_+) < 0$. Therefore $F_{a_+} \in \mathcal{LP}$. Both endpoint inequalities persist in a neighborhood by continuity, giving the strict enclosure claimed. $\square$

5 Correction of the published upper bound

The paper's Theorem 2(ii) asserts membership for every $a \geq 3.91719$. That assertion is false: Corollary 2 gives, for example, $F_{3.9642} \notin \mathcal{LP}$. The source of the discrepancy is visible in the denominator-clearing step of Lemma 5. At the chosen point $x = 2(a^2 + 1)/3$, the displayed expression in the q_j is correct, but after multiplication by the common positive denominator its fifth contribution must be

$$-96(a^2 + 1)^4(a^6 + 1),$$

not the printed $-96(a^2 + 1)^4(a^5 + 1)$. The printed expression therefore exceeds the correct one by $96a^5(a - 1)(a^2 + 1)^4$. Direct evaluation at $a = 3.91719$ gives $S_{6,a}(2(a^2 + 1)/3) \approx 0.0099216822 > 0$, rather than the required nonpositive value.

This correction does not affect Theorem 1, the sign characterization used above. It removes only the claimed 3.91719 upper bound; the monotonicity proof and exact section certificates replace it with a genuine sharp-threshold theorem and a four-decimal localization.

6 Verification and novelty scope

Run the exact, dependency-free certificate with

```
python code/exact_certificate.py
```

It uses only integer arithmetic and `fractions.Fraction`; no floating-point comparison enters either endpoint proof. The central monotonicity argument is analytic and does not depend on this computation.

A bounded search through 11 August 2026 covered the run’s result and attempt indexes and web searches for the exact title, arXiv identifier, conjecture, parameter monotonicity, and the constants 3.91719 and 3.9642. It found the original paper and related later work, but no later source resolving this conjecture or correcting Theorem 2(ii). Novelty confidence is moderate pending specialist review; mathematical confidence is high, with the main dependency made explicit above.

References

- [1] T. H. Nguyen, *On the conditions for a special entire function relative to the partial theta-function and the Euler function to belong to the Laguerre–Pólya class*, arXiv:1912.10035 (2019). <https://arxiv.org/abs/1912.10035>.